IIMSTC INTERSHIP DOCUMENT COHORT 8

Date: 26/02/2026

Mentor: Suhas Gajanana

Working hours: 10am – 1pm & 3pm – 5pm

Total Members in the Cohort: 15

Team Members Present: 11

Team Members Absent: 4 (Darshan Reddy S, Reethu G S, Poojith N Naik, Mohammed Sameer)

Objective of the day: To initiate the National GDP Growth Predictor by defining the project structure, assigning responsibilities, forming dedicated teams, and commencing the data collection process.

Work Completed: During today's session, we held a detailed discussion with our mentor regarding the scope, objectives, and structured workflow of the National GDP Growth Predictor. The complete pipeline was outlined, covering stages such as dataset collection, data preprocessing, time series alignment, exploratory data analysis, model development, and model evaluation. To ensure systematic execution and effective collaboration, the cohort was divided into five specialized teams, each responsible for a specific phase of the workflow.

Team 1 was assigned Dataset Collection (Bharath Ashok, Poojith N Naik, Bharath G V). Team 2 was assigned Data Cleaning (Aasim Pasha, Khadija Tul Kubra, Ranjan). Team 3 was assigned Time Series Alignment and Exploratory Data Analysis (Monish K, Mohammed Sameer, Reethu G S). Team 4 was assigned Model Development (Nithin Gowda, Kanish Rishab, Rashi Simpi). Team 5 was assigned Model Evaluation (Mohammed Bilal Raza, Saikeerti, Darshan Reddy). Following the role allocation, the dataset collection process was initiated by identifying relevant and reliable GDP-related sources to establish a strong foundation for subsequent stages.

Attachments:

IIMSTC INTERSHIP DOCUMENT COHORT 8

Date: 27/02/2026

Mentor: Suhas Gajanana

Working hours: 10am – 1pm & 3pm – 5pm

Total Members in the Cohort: 15

Team Members Present: 13

Team Members Absent: 2 (Reethu G S, Mohammed Bilal Raza S H)

Objective of the day: To carry out data preprocessing and ensure that the collected dataset was cleaned, structured, and prepared for further analysis and modeling.

Work Completed: D During today's session, Team 2 executed the data preprocessing phase in a structured and systematic manner. The raw dataset was cleaned by handling missing values using appropriate techniques and removing irrelevant or redundant columns. Key indicators such as GDP per capita and Internet usage were carefully selected to ensure that only meaningful variables were retained for analysis. All relevant fields were reviewed to maintain consistent and appropriate data types to prevent issues in subsequent stages.

The dataset was then restructured from wide format to long (time-series) format to make it suitable for regression and analytical modeling, ensuring that each country-year combination was clearly represented. Formatting inconsistencies were identified and resolved, and the refined dataset was exported in CSV format. Comprehensive validation and quality checks were performed, including verification of correct year extraction, duplicate removal, numerical conversion of values, and systematic sorting by country and year. Final confirmation was carried out to ensure that the dataset was clean, logically organized, and fully prepared for the upcoming analytical and modeling processes.

Doubts/Problems encountered during the Session: During the discussion, minor clarifications were required regarding the appropriate method for handling missing values and ensuring correctness in the wide-to-long transformation process. Data type consistency was also reviewed after restructuring. All concerns were discussed with the mentor and resolved during the session, and no major issues were encountered.

Attachments:

IIMSTC INTERSHIP DOCUMENT COHORT 8

Date: 28/02/2026

Mentor: Suhas Gajanana

Working hours: 10am – 1pm & 3pm – 5pm

Total Members in the Cohort: 15

Team Members Present: 10

Team Members Absent: 5 (Bharath Ashok Devadiga, Bharath G V, Mohammed Bilal Raza, Khadija Tul Kubra, Mohammed Sameer)

Objective of the day: To perform time series alignment and conduct exploratory data analysis (EDA) in order to understand trends, patterns, and relationships within the dataset and prepare it for modeling.

Work Completed: During today's session, Team 3 carried out the time series alignment and exploratory data analysis phase as assigned. The dataset was systematically sorted by country and year to ensure strict chronological consistency, which is essential for time-series modeling. Interpolation techniques were applied to handle missing values in a structured and statistically appropriate manner, ensuring continuity without introducing bias. Additional time-based features such as lag variables, growth rates, and rolling averages were engineered to capture temporal dependencies, smooth short-term fluctuations, and improve overall data stability. These enhancements strengthened the dataset and made it more suitable for predictive modeling.

In addition to alignment, exploratory data analysis was conducted to gain deeper insights into the dataset. The distribution of GDP per capita and Internet usage was analyzed across countries and time periods to observe patterns and variations. Correlation analysis was performed to understand the strength and direction of relationships between variables. Various visualizations, including scatter plots, trend lines, and correlation heatmaps, were generated to visually interpret the data and support analytical findings. These analyses provided a clearer understanding of how internet penetration trends may relate to economic growth indicators.

Finally, validation checks were conducted to ensure structural consistency and data reliability. The dataset was reviewed for duplicate entries, anomalies, and formatting inconsistencies. The correctness of newly engineered features such as lag and growth rate variables was verified to ensure accurate calculations. These steps ensured that the dataset was clean, logically organized, and fully prepared for the subsequent model development stage.

Doubts/Problems encountered during the Session: During the session, minor clarifications were required regarding the correct implementation of interpolation techniques and the interpretation of correlation results in the context of economic indicators. The team discussed these aspects with the mentor, and the concerns were resolved during the meeting. No major technical issues were encountered.

Attachments:

IIMSTC INTERSHIP DOCUMENT COHORT 8

Date: 02/03/2026

Mentor: Suhas Gajanana

Working hours: 10am – 1pm & 3pm – 5pm

Total Members in the Cohort: 15

Team Members Present: 14

Team Members Absent: 1 (Mohammed Bilal Raza S H)

Objective of the day: To develop and discuss predictive models for GDP growth by implementing and evaluating different regression techniques through an interactive session.

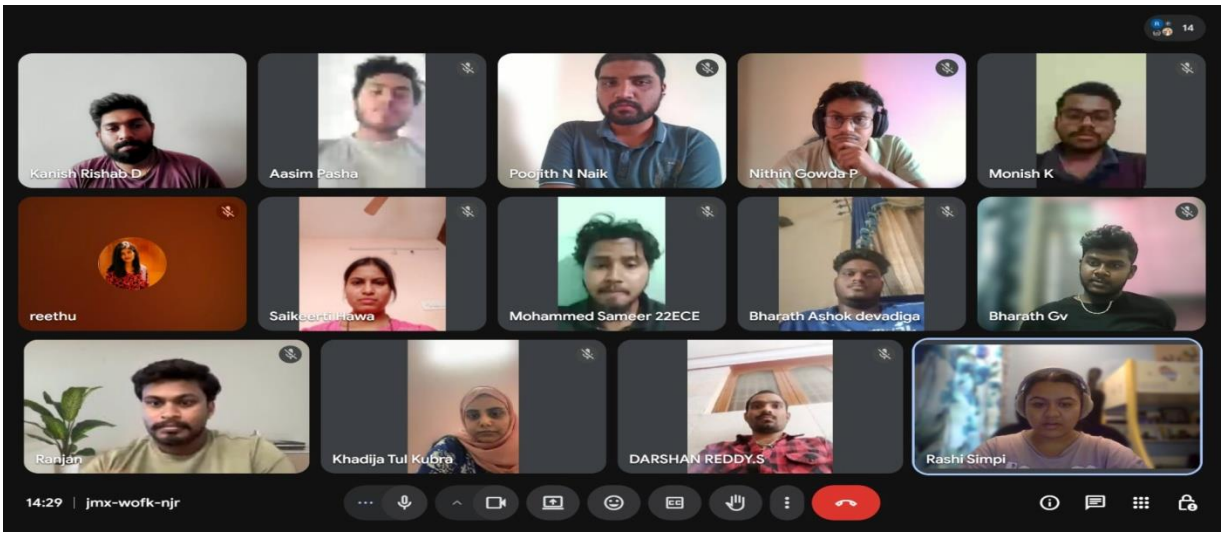
Work Completed: During today's session, an interactive discussion was conducted on model development, with a focus on selecting appropriate regression techniques to analyze the relationship between Internet Penetration Rate and GDP per Capita. The team implemented and reviewed multiple modeling approaches as per the assigned responsibilities, emphasizing interpretability, stability, and relevance to economic analysis. The session encouraged collaborative discussion on selecting suitable algorithms and understanding their applicability in real-world economic forecasting.

Linear Regression models were developed to establish a baseline understanding of the relationship between internet usage and economic growth. A basic linear regression model was implemented using raw data to determine the fundamental association between the variables. This was followed by a smoothed regression model to reduce volatility commonly observed in economic data. Additionally, a lag-based multiple linear regression model was constructed to capture economic continuity over time and incorporate temporal dependencies into the analysis.

To account for potential non-linear relationships, a Polynomial Ridge Regression model was implemented by introducing polynomial features. Ridge (L2) regularization was applied to control overfitting and address multicollinearity among the engineered features, thereby enhancing model stability. Furthermore, a Lasso Regression model was developed to support both prediction and feature selection. By applying L1 regularization, less significant feature coefficients were reduced toward zero, helping identify the most influential predictors affecting GDP per capita. These modeling approaches collectively strengthened the analytical framework and prepared the foundation for subsequent evaluation and performance assessment.

Doubts/Problems encountered during the Session: During the discussion, clarifications were required regarding the selection of appropriate regression techniques and the impact of regularization methods on model performance and interpretability. Questions were also raised about handling multicollinearity and choosing suitable polynomial degrees. These doubts were addressed during the interactive session, and all concerns were resolved with mentor guidance.

Attachments:



IIMSTC INTERSHIP DOCUMENT COHORT 8

Date: 03/03/2026

Mentor: Suhas Gajanana

Working hours: 10am – 1pm & 3pm – 5pm

Total Members in the Cohort: 15

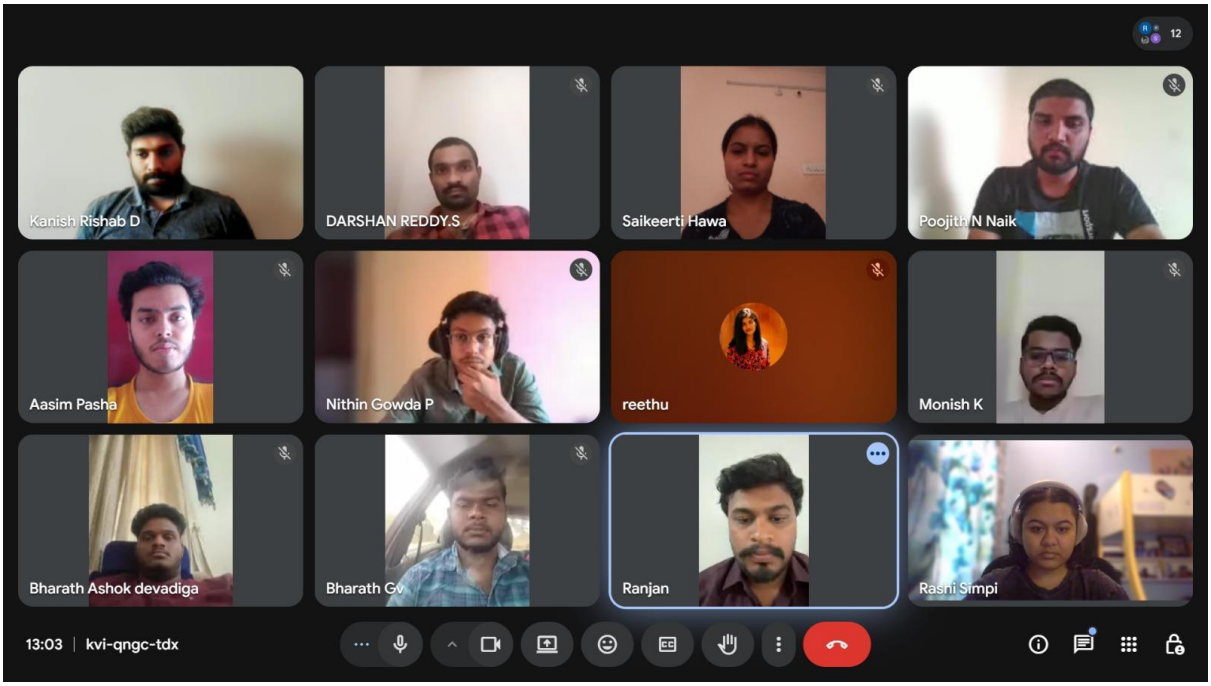
Team Members Present: 12

Team Members Absent: 3 (Mohammed Bilal Raza S, Khadija Tul Kubra, Mohammed Sameer)

Team Members Absent: To evaluate the developed regression models and assess their performance using appropriate statistical metrics and visual analysis.

Objective of the day: To initiate the National GDP Growth Predictor by defining the project structure, assigning responsibilities, forming dedicated teams, and commencing the data collection process.

Work Completed: During today's session, Team 5 carried out the model evaluation phase. The developed regression models were evaluated using performance metrics such as R^2 (coefficient of determination), RMSE (Root Mean Squared Error), and MAE (Mean Absolute Error) to assess prediction accuracy and model reliability. These metrics were compared to determine which model best captured the relationship between Internet Penetration Rate and GDP per Capita.

Graphical analysis was also performed by plotting Internet usage against GDP values along with the fitted regression curves to visually interpret model performance. Based on both statistical results and visual interpretation, the most suitable model was identified. The evaluation stage was successfully completed, marking the conclusion of the modeling workflow.

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